

Microsoft Teams Operator Connect Deployment Project

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Project summary

This project demonstrates the deployment of Microsoft Teams Operator Connect using a certified operator provider.

The project focuses on enabling PSTN calling services directly within Microsoft Teams while eliminating the complexity of traditional SBC infrastructure.

The implementation includes carrier onboarding, emergency location configuration, phone number provisioning, user assignment, and end-to-end PSTN validation.

Business Objective

Organizations adopting Microsoft Teams voice services require secure and scalable PSTN connectivity while minimizing infrastructure overhead.

This project demonstrates how Operator Connect simplifies voice deployment by allowing organizations to consume carrier services directly through Microsoft Teams.

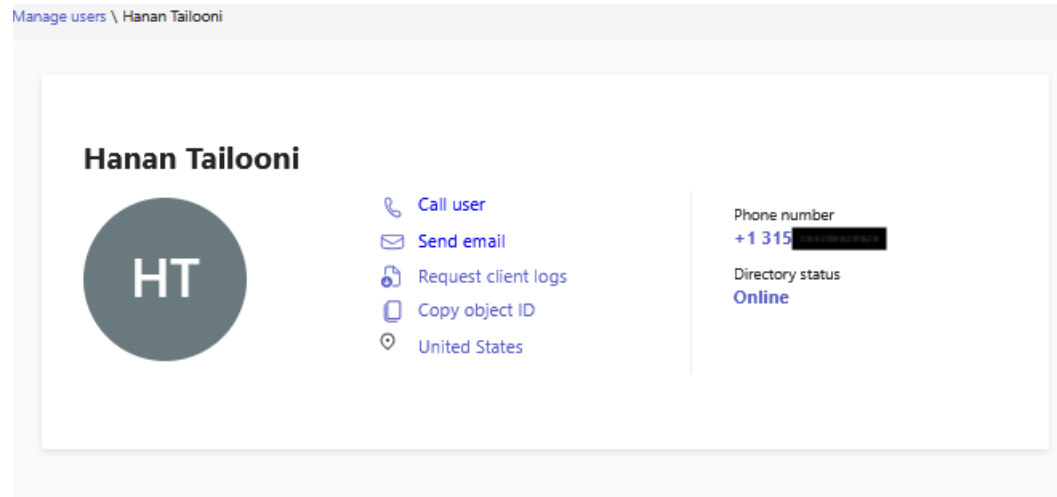
Solution Architecture

PSTN → Certified Carrier/Operator connect → Microsoft Teams Cloud → Teams users

Implementation Phases

Phase 1 – Environment preparation

- 1- Removed previous Direct Routing associations
- 2- Prepared Teams Voice environment and validated Teams licensing



- 3- Used PowerShell/Cloud shell to validate user settings and number's associations:

```
PS /home/hanan> Import-Module MicrosoftTeams
PS /home/hanan> Connect-MicrosoftTeams -UseDeviceAuthentication
To sign in, use a web browser to open the page https://login.microsoft.com/device and enter the code LBUR5TTFA to authenticate.

Account                        Environment TenantId
-----
Hanan@Tailooni.onmicrosoft.com AzureCloud 61ac7bd9-e147-4686-9c5f-ecf9af717e80 61ac7bd9-e147-4686-9c5f-ecf9af717e80

PS /home/hanan> Get-CsOnlineUser hanan@tailooni.onmicrosoft.com | fl DisplayName,EnterpriseVoiceEnabled,HostedVoiceMail

DisplayName      : Hanan Tailooni
EnterpriseVoiceEnabled : True

PS /home/hanan>
```

```
PS /home/hanan> Get-CsPhoneNumberAssignment | ft
ActivationState AssignedPstnTargetId AssignmentBlockedState AssignmentBlockedUntil AssignmentCategory Capability
-----
Activated
Activated d37f47d2-fdd4-497c-9d30-13530587ea3c NotBlocked Primary {Geographic, InboundCalling, OutboundCalling,
{Geographic, InboundCalling, OutboundCalling,
PS /home/hanan>
```

Phase 2 – Operator Connect Onboarding

1- Established relationship with our certified Operator Connect/Telnyx

Operator Connect
Manage partnerships with your phone operators. [Learn more](#)

Operator summary

113

All operators

1

My operator

All operators My operators

This page only displays the operators with whom you initiated a relationship by providing Consent. Relationships initiated by the operator by uploading telephone number may not be displayed here.

Edit Remove 1 item Search for operators

Operator	Served regions	Acquired phone numbers	Consent last updated	Updated by	Status	Offer	Contract status
Telnyx	United States	2	Jun 2, 2026	Hanan Tailooni	Active	Offer details	Agreement signed on J...

2- Completed operator onboarding workflow and synchronized carrier environment with Microsoft Teams

Microsoft Operator Connect Zoom Provider Exchange

Operator Connect ★

Filters

Type	Status	Outbound Voice Profile	SIP Connection	Date Created	Tags
Tailooni Operator Connect	●	default		Jun 02 2026	+

Phase 3 – Number Provisioning

1- Imported Existing PSTN numbers to Telnyx Operator Connect

[← Back to connections](#)

Operator Connect connection

Quickstart

SettingsNumbersNumber HistoryNotification

Added numbers

Add numbers

Filters

Phone Number	Country	Emergency Location		
+1315 [REDACTED]	US	US E911 Test	Dynamic Emergency Address	✎ 🗑
+1315 [REDACTED]	US	US E911 Test	Dynamic Emergency Address	✎ 🗑

2- Configured emergency locations in Teams and they synchronize directly to Telnyx

Emergency addresses

An emergency address is a civic address—the physical or street address of a place of business for your organization used to route emergency calls to the appropriate dispatch authorities and to assist in locating an emergency caller. You can add places and specify a floor, building, wing, or office number to give an emergency address a more exact location. [Learn more](#)

[+ Add](#) [✎ Edit](#) [🗑 Delete](#)

✓	Description	Country or region	Address	Phone numbers	Voice users	Address status ⓘ	Organization name
	E911 location	Jordan	[REDACTED]	0	0	Validated	Talooni
	US E911 Test	United States	[REDACTED]	2	1	Validated	Talooni

Phase 4 – User Assignment

1- Assigned Operator Connect numbers and validated Enterprise Voice Enablement under targeted users.

The screenshot shows the Microsoft Teams Admin Center interface. The top navigation bar includes tabs for Account, Teams, Voice, Voicemail, Meetings & calls, Apps, Teams devices, and Policies. The 'Account' tab is selected.

Assigned phone number

Primary phone number [Edit](#)

Phone number
+1 315 [REDACTED]

Phone number type
Operator Connect

[More details](#) [v](#)

Audio Conferencing [Edit](#)

Audio Conferencing
On [Send conference info in email](#)

PIN
••••• [Reset PIN](#)

Default conferencing toll phone number
none

Invites from this user can include toll-free number
Off

Dial-out permission
Any destination

```
Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help v

PS /home/hanan>
PS /home/hanan>
PS /home/hanan> Get-CsPhoneNumberAssignment -TelephoneNumber +1315 [REDACTED]

TelephoneNumber      : +1315 [REDACTED]
OperatorId           : [REDACTED]
NumberType           : OperatorConnect
ActivationState       : Activated
AssignedPstnTargetId : 
AssignmentCategory    : 
Capability            : {Geographic, InboundCalling, OutboundCalling, UserAssignment}
City                 : Hamilton
CivicAddressId        : [REDACTED]
IsoCountryCode        : US
IsoSubdivision        : New York
LocationId            : [REDACTED]
LocationUpdateSupported : True
NetworkSiteId         : 
PortInOrderStatus     : Completed
PstnAssignmentStatus  : Unassigned
PstnPartnerId         : [REDACTED]
PstnPartnerName       : Telnyx
```

Phase 5 – Validation Testing

1- Tested Outgoing calls:

▼ History

Outgoing ▾

☰



+1 [REDACTED]

📞 Outgoing

👤

12:49 PM

14s



Service provided by 

1

Total calls

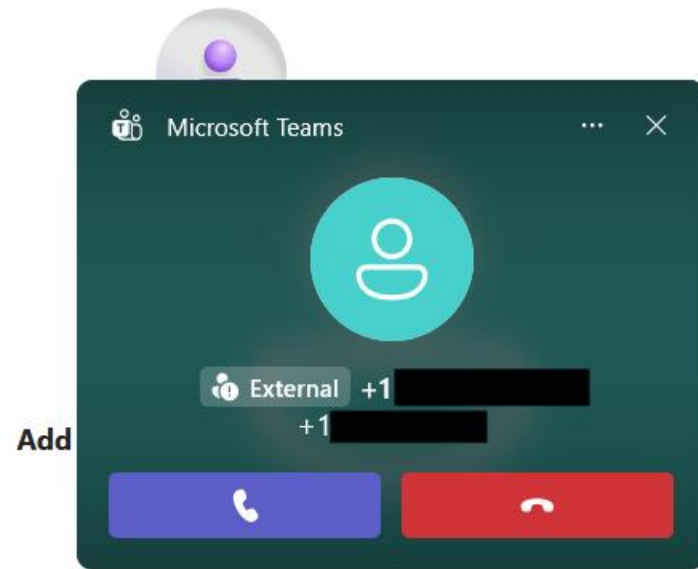
🔧 Troubleshoot

1 item

☰

Phone number	Caller ID	Operator	Call type	Destination dialed	Cost	Currency	Duration	Source of call duration	Domestic/international
.. +1 [REDACTED]	+1 315 [REDACTED]	Telnyx	📞 oc_user_out	US	0		00:00:14	Microsoft	--

2- Tested Incoming calls:



Calls

Personal

▼ History

Missed ▼



+1 [redacted] 
 Missed incoming

Report

Date range

PSTN usage

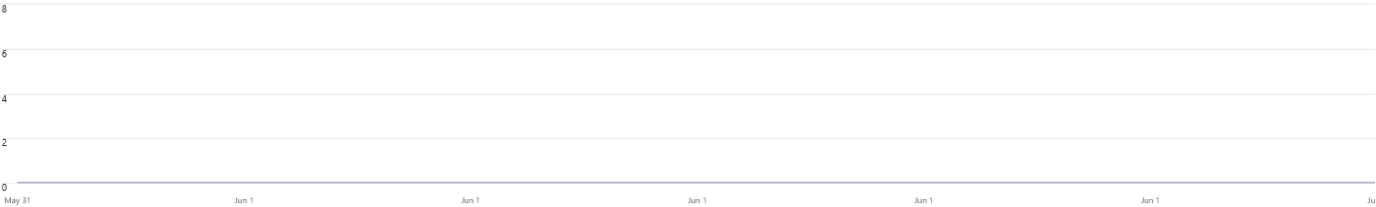
Last 24 hours

Run report

PSTN usage

Jun 02, 2026 4:56:40 PM UTC | Date range: Jun 01, 2026 - Jun 02, 2026

Calling plan & Operator Connect | Direct Routing



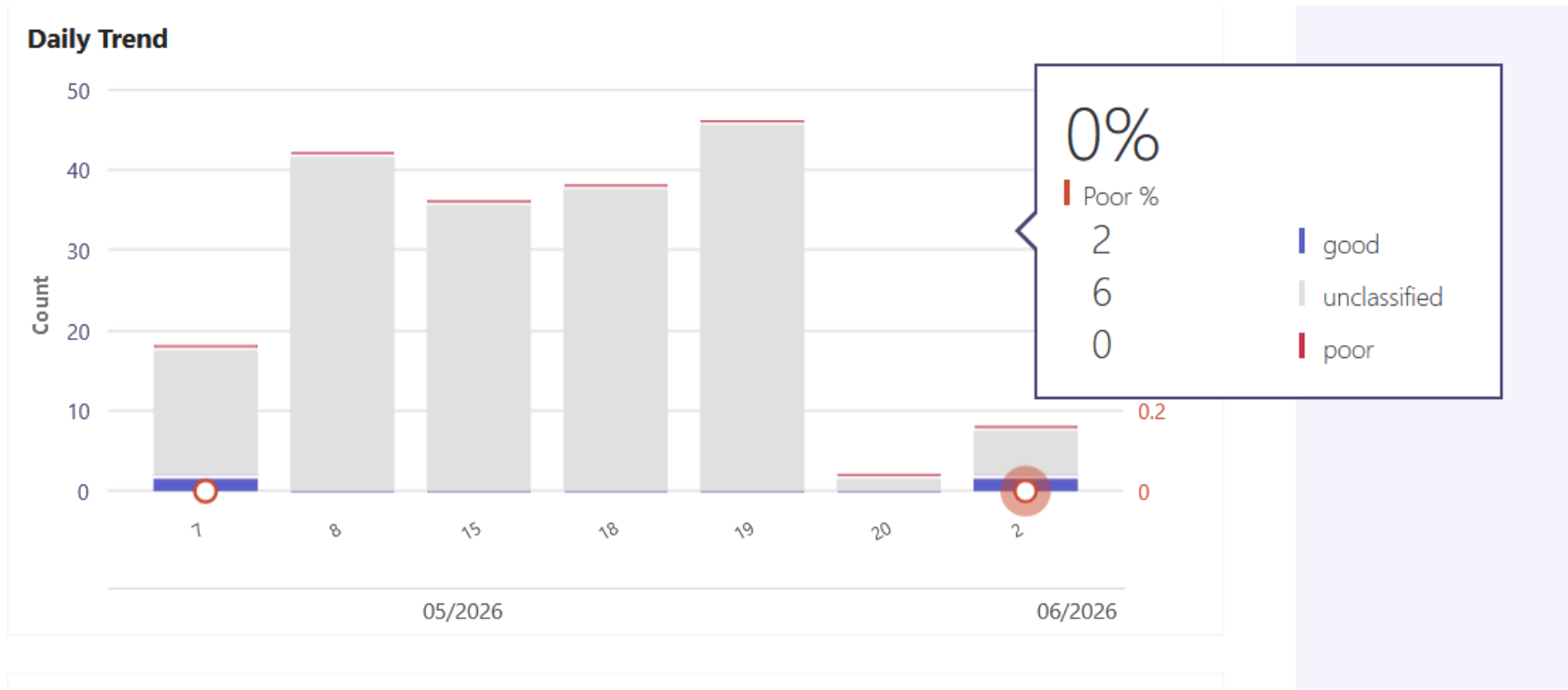
6
Total calls

Troubleshoot 6 items								
Start time (UTC) ↓	Display name	Username	Phone number	Caller ID	Operator	Call type	Destination dialed	Cost
Jun 2, 2026, 4:55:00 PM	Hanan Tallooni	Hanan	+1 315	+1	Teinyx	oc_user_in	US	0
Jun 2, 2026, 4:49:55 PM	Hanan Tallooni	Hanan@Tallooni.onmicr...	+1	+1 315	Teinyx	oc_user_out	US	0
Jun 2, 2026, 3:27:56 PM	Hanan Tallooni	Hanan@Tallooni.onmicr...	+1	+1 315	Teinyx	oc_user_out	US	0

Successful vs Unsuccessful Calls:

Two calls were successfully completed with verified two-way audio and no issues observed. The remaining six unclassified calls represent unanswered calls rather than failed calls.

Screenshot from CQD:



Technical Challenges Encountered

Challenge 1:

Direct Routing inventory conflicts

Resolution:

Removed legacy number assignments and cleaned voice inventory

Challenge 2:

Operator synchronization delays

Resolution:

Completed synchronization workflow and refreshed provider relationship

Challenge 3:

Emergency address dependencies

Resolution:

Configured emergency locations and associated location metadata

Challenge 4:

Geographic assignment mismatch

Resolution:

Aligned usage location and number geography requirements

Results

Successfully deployed Microsoft Teams Operator Connect using Telnyx while enabling full PSTN functionality.

Key Outcomes:

- Functional enterprise voice deployment
- Successful PSTN integration
- Reduced infrastructure complexity
- Validated inbound and outbound calling
- Working Teams Phone deployment

Skills Demonstrated

- Microsoft Teams Administration
- Teams Phone Deployment
- Operator Connect
- Enterprise Voice
- Teams PowerShell
- PSTN Integration
- Troubleshooting and Root Cause Analysis
- Carrier Integration

Sip signaling from Telnyx side:

